

# Documentation – Parcel Scanner Sim API Testing

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## 1. Project Description

A REST API created locally using the lightweight and asynchronous **Ktor** framework. It replaced the previously used **Postman MockServer**, which caused issues when used with methods other than GET.

## 2. API Technical Information

The API is run locally for testing purposes only, using the localhost base URL and port 8080.

Data format used: JSON.

Authorization for login endpoints uses a simple Bearer Token (JWT).

## 3. List of Endpoints

| Endpoint          | Method | Authorization |
|-------------------|--------|---------------|
| /auth/login       | POST   | Yes           |
| /history/packages | GET    | Yes           |
| /history/packages | POST   | Yes           |
| /history/forms    | GET    | Yes           |
| /history/forms    | POST   | Yes           |
| /auth/logout      | POST   | No            |

- **/auth/login**

Endpoint used for logging into our application. It is utilized when tapping an NFC card in the NFC Logon system. It checks whether the ID stored on the card exists in the system. If it does, it returns a JSON body containing the data required for further processing in the application (such as employee session data, shift start time, token, and status).

- **/history/packages**

For the GET method, we verify the logged-in employee (their ID) and then populate the JSON body with a list of packages assigned to them. The POST method is also used, this time to add a package to the list.

- **/history/forms**

For the GET method, we verify the logged-in employee (their ID) and then populate the JSON body with a list of forms and documents assigned to them. The POST method is also used, this time to add a form or document to the list.

- **/auth/logout**

Method used to log out of the application. It clears the session data and returns a response with status code 200.

## 4. Strategia testowania API

Using the test scenarios created below, we will simulate API requests to verify the correctness of the returned status codes as well as the correctness of the returned data. We will also check the system's behavior when invalid and boundary data are used. Additionally, we plan to test the performance of the requests using performance testing tools.

### **Authorization –**

TC-A01 : Login with a valid UID

TC-A02 : Login with a non-existing UID

TC-A03 : Login with an empty body

TC-A04 : Successful user logout

### **History (Packages) –**

TC-H01 : Retrieve package history for a logged-in user

TC-H02 : Retrieve package history using an invalid ID

TC-H03 : Retrieve package history without providing an ID

TC-H04 : Add a new package with valid values

TC-H05 : Add a new package with invalid values

TC-H06 : Add a new package without providing any values

### **History (Forms) –**

TC-H07 : Retrieve form history for a logged-in user

TC-H08 : Retrieve form history using an invalid ID

TC-H09 : Retrieve form history without providing an ID

TC-H10 : Add a new form with valid values

TC-H11 : Add a new form with invalid values

TC-H12 : Add a new form without providing any values

## 5. Test Cases

|                 |                                                                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-A01                                                                                                                                                                                 |
| Priority        | High                                                                                                                                                                                   |
| Description     | Verify login using a valid identifier that exists in the system.                                                                                                                       |
| Input Data      | Method: POST<br>Body: {"cardUid": "DEV_MODE_UID_123"}                                                                                                                                  |
| Steps           | 1. Send a POST request to /auth/login                                                                                                                                                  |
| Expected Result | Status code: 200 OK.<br>The body contains a User object with Jan Kowalski's data.<br>The body contains a Session object with a generated sessionID.<br>Request response time < 500 ms. |

|                 |                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-A02                                                                                                                           |
| Priority        | High                                                                                                                             |
| Description     | Verify login using an identifier that does not exist in the system.                                                              |
| Input Data      | Method: POST<br>Body: {"cardUid": "FAKE_UID_123444"}                                                                             |
| Steps           | 1. Send a POST request to /auth/login                                                                                            |
| Expected Result | Status code: 404 Not Found.<br>The body contains an error message appropriate to the request.<br>Request response time < 500 ms. |

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| Identifier      | TC-A03                                                         |
| Priority        | Medium                                                         |
| Description     | Verify login without providing any values in the request body. |
| Input Data      | Method: POST<br>Body: {}                                       |
| Steps           | 1. Send a POST request to /auth/login                          |
| Expected Result | Status code: 400 Bad Request.                                  |

|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| Identifier      | TC-A04                                                                              |
| Priority        | Medium                                                                              |
| Description     | Verify successful user logout.                                                      |
| Input Data      | Method: POST<br>Body: -                                                             |
| Steps           | 1. Send a POST request to /auth/logout                                              |
| Expected Result | Status code: 200 OK.<br>The body contains information confirming successful logout. |

|                 |                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-H01                                                                                                                                                                |
| Priority        | High                                                                                                                                                                  |
| Description     | Verify retrieval of the package list assigned to a specific logged-in user existing in the system.                                                                    |
| Input Data      | Method: GET<br>Header: X-User-ID: EMP-001                                                                                                                             |
| Steps           | 1. Send a GET request to /history/packages                                                                                                                            |
| Expected Result | Status code: 200 OK.<br>The body contains an array of Package objects.<br>The returned package list belongs to the assigned user.<br>Request response time < 1500 ms. |

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| Identifier      | TC-H02                                                                                  |
| Priority        | Medium                                                                                  |
| Description     | Verify retrieval of the package list for a user providing an invalid (non-existing) ID. |
| Input Data      | Method: GET<br>Header: X-User-ID: TEST-9901                                             |
| Steps           | 1. Send a GET request to /history/packages                                              |
| Expected Result | Status code: 401 Unauthorized                                                           |

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Identifier      | TC-H03                                                                                   |
| Priority        | Medium                                                                                   |
| Description     | Verify retrieval of the package list without providing an ID.                            |
| Input Data      | Method: GET<br>Header: -                                                                 |
| Steps           | 1. Send a GET request to /history/packages                                               |
| Expected Result | Status code: 401 Unauthorized<br>The response body returns an appropriate error message. |

|                 |                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-H04                                                                                                                                                                                                          |
| Priority        | High                                                                                                                                                                                                            |
| Description     | Verify successful saving of a scanned package in the database.                                                                                                                                                  |
| Input Data      | Method: POST<br>Header: X-User-ID: EMP-001<br>Body:<br>{<br>"trackingNumber": "TESTOWYADD-009",<br>"courier": "INPOST",<br>"scanDate": "2025-11-05",<br>"weightClass": "A",<br>"shipmentType": "DOCUMENTS"<br>} |
| Steps           | 1. Send a POST request to /history/packages                                                                                                                                                                     |
| Expected Result | Status code: 201 Created.<br>The returned object in the body contains the correctly submitted package data.<br>The package is visible in the list after retrieving it again.                                    |

|                 |                                                                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-H05                                                                                                                                                                                                       |
| Priority        | Średni                                                                                                                                                                                                       |
| Description     | Verify saving of a scanned package with data provided in an invalid format.                                                                                                                                  |
| Input Data      | Method: POST<br>Header: X-User-ID: EMP-001<br>Body:<br><pre>{   "trackingNumber": "TESTOWYADD-009",   "courier": null,   "scanDate": "2025-11-05",   "weightClass": 9,   "shipmentType": "DOCUMENTS" }</pre> |
| Steps           | 1. Send a POST request to /history/packages                                                                                                                                                                  |
| Expected Result | Status code: 400 Bad Request.<br>The response should contain an error code.                                                                                                                                  |

|                 |                                                                |
|-----------------|----------------------------------------------------------------|
| Identifier      | TC-H06                                                         |
| Priority        | Medium                                                         |
| Description     | Verify saving of a scanned package without providing any data. |
| Input Data      | Method: POST<br>Header: -                                      |
| Steps           | 1. Send a POST request to /history/packages                    |
| Expected Result | Status code: 401 Unauthorized                                  |

|                 |                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-H07                                                                                                                                                          |
| Priority        | High                                                                                                                                                            |
| Description     | Verify retrieval of the form list assigned to a specific logged-in user existing in the system.                                                                 |
| Input Data      | Method: GET<br>Header: X-User-ID: EMP001                                                                                                                        |
| Steps           | 1. Wyślij zapytanie GET na /history/forms                                                                                                                       |
| Expected Result | Status code: 200 OK.<br>The body contains an array of Form objects.<br>The returned form list belongs to the assigned user.<br>Request response time < 1500 ms. |

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| Identifier      | TC-H08                                                                               |
| Priority        | Medium                                                                               |
| Description     | Verify retrieval of the form list for a user providing an invalid (non-existing) ID. |
| Input Data      | Method: GET<br>Header: X-User-ID: TEST9901                                           |
| Steps           | 1. Send a GET request to /history/forms                                              |
| Expected Result | Status code: 401 Unauthorized                                                        |

|            |        |
|------------|--------|
| Identifier | TC-H09 |
|------------|--------|

|                 |                                                            |
|-----------------|------------------------------------------------------------|
| Priority        | High                                                       |
| Description     | Verify retrieval of the form list without providing an ID. |
| Input Data      | Method: GET<br>Header: -                                   |
| Steps           | 1. Send a GET request to /history/forms                    |
| Expected Result | Status code: 401 Unauthorized                              |

|                 |                                                                                                                                                                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-H10                                                                                                                                                                                                                                                                           |
| Priority        | High                                                                                                                                                                                                                                                                             |
| Description     | Verification of saving a correctly scanned form in the database                                                                                                                                                                                                                  |
| Input Data      | Method: POST<br>Header: X-User-ID: EMP-001<br>Body:<br><pre>{   "firstName": "Józef",   "lastName": "Testowy",   "date": "2025-10-27",   "id": "TESTFORM-001",   "address": null,   "phoneNumber": null,   "email": null,   "additionalNotes": "testowy POST formularza" }</pre> |
| Steps           | 1. Send a POST request to /history/forms                                                                                                                                                                                                                                         |
| Expected Result | Status code: 201 Created.<br>The object returned in the response body contains the correctly entered data of the submitted form.<br>The form is visible in the list after retrieving it again.                                                                                   |

|                 |                                                                                                                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identifier      | TC-H11                                                                                                                                                                                                                                                      |
| Priority        | Medium                                                                                                                                                                                                                                                      |
| Description     | Verification of saving a form with data provided in an invalid format                                                                                                                                                                                       |
| Input Data      | Method: POST<br>Header: X-User-ID: EMP-001<br>Body:<br><pre>{   "firstName": null,   "lastName": "Testowy",   "date": "2025-10-27",   "id": "TESTFORM-001",   "address": null,   "phoneNumber": 12345,   "email": false,   "additionalNotes": false }</pre> |
| Steps           | 1. Send a POST request to /history/forms                                                                                                                                                                                                                    |
| Expected Result | Status code: 400 Bad Request<br>The response should contain an error code.                                                                                                                                                                                  |



|                 |                                                          |
|-----------------|----------------------------------------------------------|
| Identifier      | TC-H12                                                   |
| Priority        | High                                                     |
| Description     | Verification of saving a form without providing any data |
| Input Data      | Method: POST<br>Header: -<br>Body:<br>{<br><br>}         |
| Steps           | 1. Send a POST request to /history/forms                 |
| Expected Result | Status code: 401 Unauthorized                            |

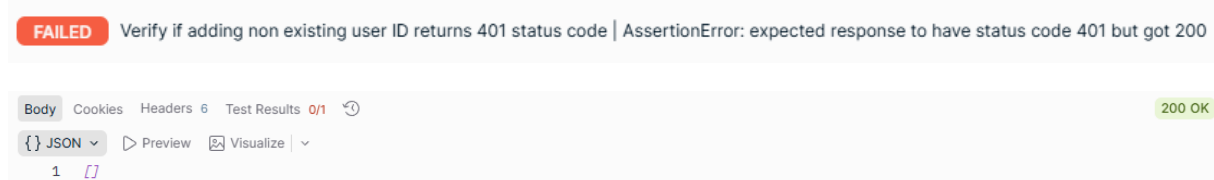
## 6. Test Execution

| Parcel Scanner Sim - API Te... - Run results                                                                                               |                                                               |            |          |           |        |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------|----------|-----------|--------|-----------------|
|  Ran today at 02:35:58 PM · <a href="#">View all runs</a> |                                                               |            |          |           |        |                 |
| Source                                                                                                                                     | Environment                                                   | Iterations | Duration | All tests | Errors | Avg. Resp. Time |
| Runner                                                                                                                                     | Parcel Scanner Sim                                            | 1          | 1s 448ms | 32        | 0      | 3 ms            |
| RUN SUMMARY                                                                                                                                |                                                               |            |          |           |        |                 |
|                                                                                                                                            |                                                               |            |          |           |        | 1               |
| > POST                                                                                                                                     | [POST] Login method without entering any credentials          | 2          | 0        |           |        |                 |
| > POST                                                                                                                                     | [POST] Login method after entering wrong credentials          | 3          | 0        |           |        |                 |
| > POST                                                                                                                                     | [POST] Login method after entering existing credentials       | 4          | 0        |           |        |                 |
| > POST                                                                                                                                     | Logout method                                                 | 2          | 0        |           |        |                 |
| ✓ GET                                                                                                                                      | [GET] Get packages history for non-existing user              | 0          | 1        |           |        |                 |
| FAIL                                                                                                                                       | Verify if adding non existing user ID returns 401 status code |            |          |           |        |                 |
| > GET                                                                                                                                      | [GET] Get packages history without entering ID                | 2          | 0        |           |        |                 |
| > GET                                                                                                                                      | [GET] Get packages history for existing user with correct ID  | 3          | 0        |           |        |                 |
| > POST                                                                                                                                     | [POST] Add package with correctly entered informations        | 2          | 0        |           |        |                 |
| > POST                                                                                                                                     | [POST] Add package with wrong types entered into body         | 2          | 0        |           |        |                 |
| > POST                                                                                                                                     | [POST] Add package without informations and ID                | 1          | 0        |           |        |                 |
| > GET                                                                                                                                      | [GET] Get forms history for existing user with correct ID     | 3          | 0        |           |        |                 |
| > GET                                                                                                                                      | [GET] Get forms history for user without entering ID          | 2          | 0        |           |        |                 |
| ✓ GET                                                                                                                                      | [GET] Get forms history for non-existing user                 | 0          | 1        |           |        |                 |
| FAIL                                                                                                                                       | Verify if adding non existing user ID returns 401 status code |            |          |           |        |                 |
| > POST                                                                                                                                     | [POST] Add form with correct informations                     | 2          | 0        |           |        |                 |
| > POST                                                                                                                                     | [POST] Add form with wrong types entered into body            | 2          | 0        |           |        |                 |
| POST                                                                                                                                       | [POST] Add form without entering body informations            | 0          | 0        |           |        |                 |

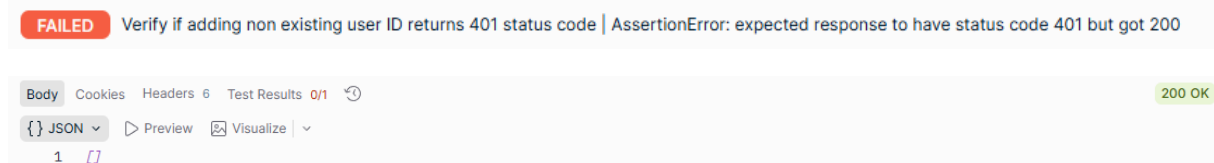
During the execution of the test suite, for each test case we could observe that two tests did not pass successfully. Therefore, after the initial analysis, two bug reports were created for the issues encountered in the API.

## 7. Defects and Fixes

|                    |                                                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title              | [BUG-01] Unhandled error when retrieving the package list for a non-existent user                                                                                             |
| Description        | When retrieving the package list by providing an Identifier in the correct format, but for a non-existent user, an empty array is returned instead of an error.               |
| Steps to Reproduce | <ol style="list-style-type: none"><li>1. In the GET request for the /history/packages endpoint, set the X-User-ID header to NULL-9901.</li><li>2. Send the request.</li></ol> |
| Expected Result    | The API response should return a 401 error code along with an appropriate error message.                                                                                      |
| Actual Result      | The API response returns status 200 OK, and the response body contains an empty JSON array.                                                                                   |



|                    |                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title              | [BUG-02] Unhandled error when retrieving the form list for a non-existent user                                                                                               |
| Description        | When retrieving the list of forms by providing an Identifier in the correct format, but for a non-existent user, an empty array is returned instead of an error.             |
| Steps to Reproduce | <ol style="list-style-type: none"><li>1. In the GET request for the /history/forms endpoint, set the X-User-ID header to TEST-000991.</li><li>2. Send the request.</li></ol> |
| Expected Result    | The API response should return a 401 error code along with an appropriate error message.                                                                                     |
| Actual Result      | The API response returns status 200 OK, and the response body contains an empty JSON array.                                                                                  |



The solution to these issues involved fixing the condition responsible for validating the provided ID and verifying whether the user exists in the list of

trusted Identifiers stored in the database.

```
// === ENDPOINT: HISTORIA PACZEK (GET) ===
get(path = "/history/packages") {

    //GET USER ID
    val userId = call.request.headers["X-User-ID"]

    if (userId == null) {
        call.respond(status = HttpStatusCode.Unauthorized, message = mapOf("error" to "Missing User ID"))
        return@get
    }

    val userPackages = Database.packagesHistory[userId] ?: emptyList()
    call.respond(status = HttpStatusCode.OK, message = userPackages)
}
```

The code snippet below shows the endpoint responsible for returning the package list. As shown, it only checks whether the `userId` is empty (i.e., `null`). If an invalid value is provided, this condition is skipped, which results in the reported issue.

```
val userExists = Database.authorizedUsers.values.any { it.id == userId }

if (!userExists) {
    call.respond(HttpStatusCode.Unauthorized, mapOf("error" to "User not authorized"))
    return@get
}
```

The solution was to properly verify whether the user exists in the list of trusted Identifiers and handle the request accordingly. The same validation was added for the second reported issue in the endpoint responsible for retrieving the list of forms.

As a result of these fixes, the test suite completed successfully, as shown in the screenshot below.

 Ran today at 03:48:54 PM · [View all runs](#)

| Source | Environment        | Iterations | Duration | All tests | Errors | Avg. Resp. Time |
|--------|--------------------|------------|----------|-----------|--------|-----------------|
| Runner | Parcel Scanner Sim | 1          | 2s 545ms | 33        | 0      | 20 ms           |

## 8. Performance Testing

In the test plan, I divided performance testing into two groups: performance testing of the application as a whole and performance testing of the API in order to examine how the endpoints handle a large number of users as well as different network conditions and latencies.

Test Cases -

| Test ID | Scenario Description                                             | Number of Users  | Ramp-up period | Loop | Verification Objectives                                                                     |
|---------|------------------------------------------------------------------|------------------|----------------|------|---------------------------------------------------------------------------------------------|
| PT-01   | Retrieving the package list under standard load                  | 15 użytkowników  | 5 seconds      | 10   | Verify that package retrieval works under a typical API usage scenario.                     |
| PT-02   | Retrieving the package list under increased traffic              | 50 użytkowników  | 10 seconds     | 20   | Verify package list retrieval under increased traffic.                                      |
| PT-03   | Retrieving the package list for a large number of users          | 125 użytkowników | 3 seconds      | 50   | Verify how the system behaves under very high traffic while retrieving the package list.    |
| PT-04   | Access to unauthorized resources (package list)                  | 100 użytkowników | 1 second       | 10   | Verify that the security mechanisms work correctly under load.                              |
| PT-05   | Retrieving the form list under standard load                     | 15 użytkowników  | 5 seconds      | 10   | Verify that retrieving the form list works correctly through the API.                       |
| PT-06   | Retrieving the form list under increased traffic                 | 50 użytkowników  | 10 seconds     | 20   | Verify form list retrieval under increased traffic.                                         |
| PT-07   | Retrieving the form list for a large number of users             | 125 użytkowników | 3 seconds      | 50   | Verify how the system behaves under very high traffic while retrieving the form list.       |
| PT-08   | Adding a very large number of packages in a short period of time | 50 users         | 5 seconds      | 10   | Verify that adding a large number of packages within a short period completes successfully. |
| PT-09   | Adding a very large number of forms in a short period of time    | 50 users         | 5 seconds      | 10   | Verify that adding a large number of forms within a short period                            |

|       |                                                         |          |           |    |                                                                                                      |
|-------|---------------------------------------------------------|----------|-----------|----|------------------------------------------------------------------------------------------------------|
|       |                                                         |          |           |    | completes successfully.                                                                              |
| PT-10 | Adding a package with invalid field types by many users | 35 users | 2 seconds | 10 | Verify that the field validation mechanism in the POST methods works correctly under high user load. |

Below are the tables from the first performance tests of our API. It can be observed that we achieved a fairly acceptable response time from the backend. The throughput for the POST methods is also not concerning. However, in some test groups it can be noticed that the percentage of failed tests ranges from **5%** to 19%, which drew my attention.

| Label                                                         | # Samples ↑ | Average | Min | Max   | Std. Dev. | Error % | Throughput | Received KB/sec | Sent KB/sec | Avg. Bytes |
|---------------------------------------------------------------|-------------|---------|-----|-------|-----------|---------|------------|-----------------|-------------|------------|
| [GET] Get package list under small network traffic            | 96          | 39      | 1   | 192   | 40.38     | 20.83%  | 17.6/sec   | 4615.95         | 2.61        | 268488.0   |
| [GET] Get forms list under small network traffic              | 96          | 48      | 1   | 471   | 63.83     | 20.83%  | 17.5/sec   | 5769.65         | 2.55        | 337193.8   |
| [POST] Adding packages with wrong types of data in short time | 350         | 13      | 0   | 183   | 20.79     | 100.00% | 149.8/sec  | 36.56           | 51.45       | 250.0      |
| [GET] Get forms list under higher network traffic             | 460         | 220     | 0   | 44457 | 2665.51   | 15.87%  | 6.9/sec    | 1954.15         | 1.02        | 291634.9   |
| [GET] Get package list under higher network traffic           | 469         | 454     | 0   | 47407 | 3826.92   | 14.29%  | 7.0/sec    | 2034.60         | 1.03        | 297824.3   |
| [POST] Correctly add form under higher network traffic        | 483         | 36      | 0   | 4125  | 191.29    | 0.00%   | 8.2/sec    | 2.63            | 3.32        | 329.6      |
| [POST] Correctly add package under higher network traffic     | 486         | 47      | 0   | 2940  | 190.62    | 0.00%   | 58.8/sec   | 17.09           | 21.05       | 297.8      |
| [GET] Try to get package list as unauthorized user            | 3849        | 50      | 0   | 31835 | 595.57    | 100.00% | 57.4/sec   | 11.72           | 8.69        | 209.0      |
| [GET] Get form list under stress testing army group           | 3869        | 207     | 1   | 49772 | 2592.86   | 15.33%  | 57.7/sec   | 16248.69        | 8.53        | 288237.2   |
| [GET] Get package list under stress testing army group        | 3902        | 181     | 0   | 49590 | 2327.95   | 17.35%  | 58.2/sec   | 15977.04        | 8.62        | 281025.6   |
| TOTAL                                                         | 14060       | 147     | 0   | 49772 | 2045.29   | 40.18%  | 209.7/sec  | 37065.49        | 35.62       | 181004.6   |

By analyzing the Ktor logs, we can observe that most of these failures returned the **500 Internal Server Error** status code, while IntelliJ reported a **ConcurrentModificationException**.

|                    |                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title              | [BUG-PERF-01] Race Condition occurring during simultaneous data access using the GET method                                                                                                                       |
| Description        | During load testing (Stress Test, 125 users), HTTP 500 Internal Server Error responses were observed for 3–10% of GET requests.                                                                                   |
| Steps to reproduce | <ol style="list-style-type: none"> <li>1. Open JMeter with the prepared test scenario loaded.</li> <li>2. Run all tests for 10, 50, and 125 users.</li> <li>3. In the last tab, select Summary Report.</li> </ol> |
| Expected Result    | All performance tests should have an Error percentage lower than 0.2%, except for negative test scenarios.                                                                                                        |
| Actual Result      | For the GET methods, the Error percentage ranges from 3% to 10%. Some GET requests return the 500 Internal Server Error status code together with a ConcurrentModificationException.                              |

To eliminate race conditions in the GET methods, different data structures were introduced. MutableMapOf and MutableList were replaced with ConcurrentHashMap and CopyOnWriteArrayList, respectively. Retesting confirmed the effectiveness of the implemented changes.

| Label                                                  | # Samples | Average | Min | Max | Std. Dev. | Error % | Throughput | Received KB/sec | Sent KB/sec | Avg. Bytes |
|--------------------------------------------------------|-----------|---------|-----|-----|-----------|---------|------------|-----------------|-------------|------------|
| [GET] Get package list under higher network traffic    | 1000      | 1       | 0   | 269 | 9.63      | 0.00%   | 101.8/sec  | 4702.01         | 15.12       | 47281.9    |
| [GET] Get package list under stress testing army group | 6250      | 2       | 0   | 253 | 8.74      | 0.00%   | 2062.7/sec | 43919.24        | 306.18      | 21803.1    |
| [GET] Get package list under small network traffic     | 100       | 4       | 0   | 270 | 26.81     | 0.00%   | 22.2/sec   | 631.68          | 3.29        | 29191.8    |
| [POST] Adding packages with wrong types of data in ... | 350       | 7       | 0   | 257 | 22.04     | 100.00% | 179.8/sec  | 43.89           | 61.75       | 250.0      |
| [POST] Correctly add package under higher network t... | 500       | 3       | 0   | 264 | 16.40     | 0.00%   | 102.0/sec  | 29.65           | 36.52       | 297.8      |
| [GET] Get form list under stress testing army group    | 6250      | 2       | 0   | 136 | 6.37      | 0.00%   | 2250.6/sec | 47988.36        | 334.08      | 21833.9    |
| [GET] Get forms list under higher network traffic      | 1000      | 1       | 0   | 62  | 3.67      | 0.00%   | 104.7/sec  | 4836.67         | 15.54       | 47303.7    |
| [GET] Try to get package list as unauthorized user     | 6250      | 1       | 0   | 92  | 5.35      | 100.00% | 2253.9/sec | 460.02          | 341.16      | 209.0      |
| [POST] Correctly add form under higher network traffic | 500       | 2       | 0   | 131 | 10.58     | 0.00%   | 107.7/sec  | 34.67           | 43.83       | 329.6      |
| [GET] Get forms list under small network traffic       | 100       | 3       | 0   | 85  | 8.98      | 0.00%   | 23.6/sec   | 836.85          | 3.43        | 36385.6    |
| TOTAL                                                  | 22300     | 2       | 0   | 270 | 8.13      | 29.60%  | 2270.9/sec | 37350.22        | 369.71      | 16842.2    |